

ALBANIA AI 2035

Albania as the Western Balkans' Next Technology Export Powerhouse

An Academic Policy Framework for Artificial Intelligence, Innovation, Talent and Technology Exports



Figure 1. Supplied first-attachment strategy dashboard.

Abstract

This paper develops the nine-part strategy presented in the supplied Albania AI 2035 dashboard into an academic policy framework. It argues that Albania's AI opportunity should be approached as an ecosystem problem linking talent, digital infrastructure, applied research, finance, industry demand, governance and export capability. The paper proposes a progression from digital adoption to digital capability, innovation, products and intellectual property, and ultimately exports. It further develops a national AI and innovation hub, a regional network, priority application domains, a stage-gated implementation roadmap and a measurement model. The framework incorporates recent evidence from the European Commission, OECD, World Bank and UNESCO. Quantitative targets retained from the original dashboard are explicitly treated as illustrative planning scenarios rather than forecasts. The paper also places human critical thinking and oversight at the center of AI deployment, consistent with UNESCO's ethical framework and the EU AI Act.

Keywords: Albania; artificial intelligence; innovation policy; technology exports; digital transformation; startups; human capital; Western Balkans; AI governance; youth retention.

1. Research framing and methodology

The starting point is the supplied first attachment and the accompanying project article. The nine numbered sections and their terminology are retained. The academic development adds recent external evidence and distinguishes source-derived propositions, evidence reported by current institutional sources, and analytical proposals. Several numerical values in the original dashboard are explicitly illustrative.

Reference currency: sources were checked against material available through September 2026. The official Albanian consultation record shows a proposed National AI Strategy for 2025–2030 was posted in August 2025 and closed in September 2025; a separate draft AI law consultation was posted in May 2026. These records should not be treated as proof that every proposed measure had already been fully adopted or implemented.

1. Introduction and Vision

Albania's next stage of digital development can be understood as a transition from technology consumption toward technology creation, commercialization, and export. The supplied first attachment expresses this ambition through six connected themes: AI innovation, young talent, regional leadership, economic growth, an innovation hub, and global impact. In academic terms, the proposed strategy is an ecosystem intervention: it seeks to connect human capital, infrastructure, research, finance, demand, institutions, and international market access rather than treating artificial intelligence as a stand-alone technology policy. This framing is consistent with recent international evidence. The World Bank's 2025 Digital Progress and Trends Report argues that AI participation depends on foundational conditions including connectivity, compute, context/data, and competency/skills. The report describes these as the “four Cs” required for inclusive AI ecosystems. For Albania, this implies that a technology-export strategy should not begin with models alone; it should begin with the institutional and productive capabilities that allow firms, universities and public organizations to use, adapt and ultimately create AI-enabled solutions.

Policy and evidence development

The policy opportunity is strengthened by Albania's existing innovation-policy architecture. Albania's National Strategy for the Development of Innovative Entrepreneurship 2024–2030 identifies the innovation/startup ecosystem, strategic sectors and inclusive development as major pillars. The European Commission's 2025 Albania Report and subsequent Growth Plan implementation provide a broader European policy context, while the European Digital Innovation Hub network creates an existing institutional mechanism for technology adoption and experimentation. The strategy proposed here should therefore be additive: it should coordinate and commercialize existing capacities rather than duplicate them.

Academic interpretation

Academic proposition: Albania's comparative opportunity is not necessarily to compete with the largest AI economies in frontier-model scale. A more testable proposition is to build specialized, export-oriented capabilities around applied AI, low-resource language technologies, sector-specific software, engineering services and digital public-sector solutions.

Evidence quotation: The World Bank describes the foundational requirements as the “four Cs”—connectivity, compute, context (data), and competency (skills).

2. From Digital Adoption to Technology Exports

The first attachment proposes a progression: Digital Adoption → Digital Capability → Innovation → Products & IP → Exports. This sequence provides a useful analytical distinction between using technology and producing economically valuable technology. Adoption can improve efficiency, but export capability requires additional capabilities: product management, intellectual property, standards, international sales, customer validation, financing and organizational learning. The OECD's 2025 work on SME AI adoption identifies connectivity, data/algorithms/compute, skills and finance as important enablers, while noting that AI adoption among SMEs remains uneven. Its 2026 D4SME survey similarly reports that off-the-shelf AI use is increasing, but strategic, targeted and secure integration into business operations remains uneven. These findings support a policy distinction between superficial adoption and organizational capability.

Policy and evidence development

The strategic implication is a stage-gate model. Public support should move organizations through clearly defined stages: establish digital processes; build internal data and technical capability; identify a measurable problem; prototype a solution; validate it with a real user; convert validated work into a repeatable product or service; and develop international market access. Each stage should have different performance indicators. Training completion is an input or activity measure; a validated prototype is an output; an international customer and recurring export revenue are outcome measures.

Academic interpretation

The central research question becomes: under what conditions does AI adoption generate new tradable products and services rather than simply increasing dependence on imported platforms? This question should guide evaluation of the strategy.

Evidence quotation: OECD research reports that “strategic, targeted and secure integration within business operations remains uneven” among surveyed SMEs.

3. National AI and Innovation Hub

The proposed National AI & Innovation Hub is conceived as a coordinating platform rather than merely a physical building. Its functions would include applied research, graduate training, startup incubation, government innovation, industry collaboration, technology testing, investment preparation and international partnerships. The first attachment represents the pipeline as Idea → Research/Prototype → Pilot → Startup/Product → Export. This proposal aligns with the existing AI4DigiT European Digital Innovation Hub. The European Commission's EDIH catalogue describes AI4DigiT as a national initiative founded by the University of Tirana, the National Agency for Information Society and the Chamber of Commerce and Industry in Tirana, with an emphasis on AI-driven transformation, funding access and low-resource languages, particularly Albanian. The appropriate policy question is therefore how a national strategy can strengthen coordination, commercialization and regional reach around existing assets.

Policy and evidence development

The hub should operate with measurable service lines: challenge discovery; research and data support; prototyping; testing; regulatory and ethical review; procurement/pilot facilitation; investment readiness; and export preparation. A portfolio approach would allow projects to enter at different maturity levels. Governance should include universities, industry, government, investors and civil society, with conflict-of-interest rules and transparent project selection.

Academic interpretation

A research-oriented hub should also generate evidence about what works. Every major program should collect comparable data on project duration, technical performance, adoption, investment, jobs, revenue, exports and survival. This converts the hub from an event-based innovation center into an evidence-generating institution.

Academic note: The propositions in this section should be treated as policy hypotheses to be tested through implementation and evaluation.

4. Regional Innovation Centers

The first attachment identifies Tirana as an anchor and proposes connections to Durrës, Shkodër, Elbasan, Vlorë and Korçë. A distributed model can reduce geographic concentration while allowing specialized experimentation. Regional centers should not be treated as miniature copies of a national hub; instead, they should focus on locally relevant productive systems and connect to a national technical and commercialization backbone.

Policy and evidence development

Potential specialization is illustrative rather than predetermined: manufacturing and industrial AI in Elbasan; logistics and tourism in Durrës; agriculture and environmental applications in Shkodër; maritime and tourism applications in Vlorë; agriculture, food and regional services in Korçë; and cross-sector AI, research, finance and national coordination in Tirana. Specializations should ultimately be selected using evidence on firms, universities, infrastructure, labor-market demand and export potential.

Academic interpretation

The regional model also creates a research opportunity: compare whether sector-focused regional ecosystems produce different rates of prototype validation, startup formation, investment or retention than general-purpose programs. Such comparative evaluation would improve policy design over time.

Academic note: The propositions in this section should be treated as policy hypotheses to be tested through implementation and evaluation.

5. Key AI Export Sectors

The first attachment identifies six priority domains: manufacturing, tourism, agriculture, government, cybersecurity and Albanian-language AI. These domains should be understood as hypotheses for export specialization rather than guaranteed winners. Their selection is defensible because they combine identifiable operational problems with potential for software or knowledge-intensive services to be reused across markets.

Policy and evidence development

Manufacturing applications may include predictive maintenance, computer vision and production optimization. Tourism applications may include demand forecasting, multilingual assistance and destination analytics. Agricultural applications may include crop monitoring, disease detection and precision irrigation. Government applications may include document classification, workflow optimization and analytical decision support. Cybersecurity applications may include threat detection and compliance support. Albanian-language AI can address speech, translation, document analysis and educational applications for a relatively under-resourced language context. AI4DigiT's stated emphasis on Albanian and other low-resource-language applications provides an institutional point of reference for the latter domain.

Academic interpretation

The academic test for sector selection should be exportability. A proposed solution should be assessed for transferability across firms, countries, languages, regulatory environments and technical infrastructures. This requires market research and customer discovery alongside engineering.

Academic note: The propositions in this section should be treated as policy hypotheses to be tested through implementation and evaluation.

6. Data, Analysis and Targets

The first attachment contains an illustrative AI Innovation Ecosystem Index and a 2030–2035 performance table. The original source explicitly characterizes these numbers as illustrative planning scenarios. An academic version should preserve that distinction because a scenario is not a forecast and should not be presented as an observed national statistic. The proposed indicators include advanced-tech graduate participation, technology startups, exporting technology firms, university–industry AI projects and private technology investment. These are useful dimensions, but they require operational definitions. For example, “technology startup” should specify whether it means newly registered firms, firms receiving investment, firms with technology as a core product, or firms meeting a revenue or employment threshold.

Policy and evidence development

A rigorous monitoring framework should include baseline, target, data source, collection frequency, responsible institution, population, denominator and verification method. Indicators should also be separated into inputs, activities, outputs, outcomes and impacts. This reduces the risk that a large number of training participants is interpreted as evidence of increased export competitiveness.

Academic interpretation

The World Bank’s 2025 framework is useful here because it emphasizes foundational AI conditions and the uneven distribution of innovation, compute, skills and data. OECD evidence adds the importance of firm-level adoption conditions. The Albanian dashboard can therefore combine ecosystem measures with firm-level evidence and international benchmarking.

Academic note: The propositions in this section should be treated as policy hypotheses to be tested through implementation and evaluation.

7. Implementation Roadmap

The first attachment proposes a five-year sequence: Year 1 governance and baseline; Year 2 experimentation and laboratories; Year 3 commercialization; Year 4 internationalization; Year 5 scale and evaluation. This sequencing can be strengthened by treating each year as a portfolio stage rather than a fixed list of activities.

Policy and evidence development

Year 1 should establish governance, baseline measurement, program rules, priority sectors and the initial fellowship/challenge mechanisms. Year 2 should emphasize prototypes and pilots. Year 3 should introduce stronger commercialization criteria and private-finance participation. Year 4 should focus on international customers, foreign R&D; relationships and European partnerships. Year 5 should conduct independent evaluation and reallocate resources toward mechanisms with demonstrated outcomes.

Academic interpretation

A critical governance principle is learning. Programs should have explicit stop, redesign and scale decisions. Public funding should become progressively more contingent on evidence as projects move from experimentation toward commercialization.

Academic note: The propositions in this section should be treated as policy hypotheses to be tested through implementation and evaluation.

8. Analytical Model and Human Critical Thinking

The first attachment presents an analytical model in which AI competitiveness is conceptualized as Talent × Infrastructure × Research × Capital × Industry Demand × Export Capability × Governance. The multiplicative form is best understood as a diagnostic heuristic: it highlights the possibility that severe weakness in one component can constrain the system even when other components are strong. It is not, by itself, a validated econometric production function.

Policy and evidence development

A Python implementation could create a transparent monitoring dashboard, calculate normalized indicators, visualize trends and conduct scenario analysis. It should preserve methodological transparency: definitions, missing data, normalization rules, weights and uncertainty should be documented. Where causal claims are made, the strategy should use appropriate empirical methods rather than relying on a composite index alone.

Academic interpretation

The critical-thinking principle is central to the academic version. UNESCO's Recommendation on the Ethics of Artificial Intelligence states that human oversight should prevent AI systems from displacing “ultimate human responsibility and accountability.” The EU AI Act likewise requires effective human oversight for high-risk systems and explicitly recognizes the risk of over-reliance on AI outputs. Therefore, AI education should include problem formulation, evidence assessment, model validation, error detection, interpretation and human review—not only tool operation.

A useful educational sequence is: define the problem → gather evidence → formulate hypotheses → analyze the process/data → use AI appropriately → validate outputs → identify uncertainty and errors → make a documented human decision → measure the result.

Evidence quotation: UNESCO states that AI systems should not displace “ultimate human responsibility and accountability.”

9. Conclusion

The first attachment's conclusion can be developed into a coherent research proposition: Albania can use existing digital infrastructure, innovation institutions, universities, EDIHs, diaspora networks and European integration mechanisms to develop an applied-AI ecosystem oriented toward exportable products and services. The strategy does not require Albania to replicate a frontier AI economy. It requires a disciplined pathway from local problems to validated solutions and then to regional and European markets.

Policy and evidence development

The proposal should be evaluated through outcomes: productivity improvements, new products, intellectual property, private investment, technology-export revenue, international customers, research outputs, graduate retention and the survival of supported ventures. The strategy should also be evaluated for distributional effects, including regional participation, inclusion and access to advanced skills.

Academic interpretation

The resulting proposition is: Build in Albania. Test in the Western Balkans. Scale into Europe. In academic terms, this is a hypothesis to be tested through measurable programs and longitudinal evaluation, not a predetermined outcome.

Academic note: The propositions in this section should be treated as policy hypotheses to be tested through implementation and evaluation.

10. Integrated conceptual framework

The nine sections can be represented as a capability chain. Talent and infrastructure create the conditions for applied research; research and problem discovery create prototypes; pilots create evidence; evidence supports commercialization; commercialization creates products and services; and export capability converts validated solutions into international economic activity. Governance operates across the chain, while measurement provides feedback.

Stage	Primary question	Evidence	Key indicator
Foundations	Are enabling conditions present?	Connectivity, data, compute, skills	Capability baseline
Innovation	Can a real problem be solved?	Prototype and validation evidence	Validated prototypes
Commercialization	Will users pay or adopt?	Customers, investment, repeatability	Revenue / adoption
Export	Can the solution travel?	International customers and compliance	Export revenue
Impact	Does it improve outcomes?	Productivity, jobs, retention	Outcome measures

11. Measurement and evaluation design

Each strategic target should be accompanied by a baseline, operational definition, source, collection frequency, responsible institution and verification procedure. Evaluation should combine descriptive dashboards with project-level evidence and, where feasible, quasi-experimental or longitudinal methods. Composite indices may be useful for communication, but they should not substitute for analysis of individual components.

12. Policy limitations and research agenda

- AI export performance cannot be inferred from training participation alone.
- The illustrative 2030–2035 targets require official baselines and feasibility testing before adoption.
- An ecosystem index involves methodological choices about normalization and weighting; sensitivity analysis is required.
- Sector prioritization should be tested against firm-level demand, labor-market evidence, infrastructure and export data.
- Human oversight and accountability need to be integrated into procurement, education and deployment standards.
- Longitudinal evaluation is needed to determine whether public support generates additional investment, exports and durable firms.

13. Selected direct quotations

World Bank (2025): “four Cs”—connectivity, compute, context (data), and competency (skills).

OECD (2026): “strategic, targeted and secure integration within business operations remains uneven.”

UNESCO (2021): AI systems should not displace “ultimate human responsibility and accountability.”

European Union, AI Act (2024): High-risk AI systems must be designed so they can be “effectively overseen by natural persons.”

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